

System Testing Checklist Report

BizTrack, business permit processing for the City of Malabon. Response to the 25-item tester checklist, the work done against it, and what still needs a decision from the LGU.

Date 28 July 2026	Repository antuK3n/biztrack	Pull requests 14	Tests 121 passing, 424 assertions
Changed 86 files			

1. How this was checked

Every item was driven through a real browser against the running application before anything was written about it. That matters here for a specific reason: an earlier round of this work was reported as complete on the strength of a TypeScript check alone, and several of those claims turned out to be wrong when someone actually clicked the buttons.

That earlier gap had a concrete cause, documented in section 6: the type-check command everyone was running checks nothing at all.

Verification used, in order of how much they prove:

- **Browser walkthrough** of each affected flow, with the database inspected afterwards to confirm what was actually persisted rather than what the screen claimed.
- **121 backend tests** (Pest), covering validation, authorization, fee computation and the workflow lifecycle.
- **Production build** (`npm run build`), which is what genuinely type-checks the frontend.

2. Checklist status, all 25 items

Eight items were already fixed before this round. They are marked as such and were re-verified rather than re-claimed as new work.

#	ITEM	STATUS	DETAIL
1	Edit profile not working	Already fixed	Changed a name, confirmed it persisted to the database.
2	Form does not match the real form	Partly blocked	Four missing fields added. Closing it fully needs the paper form. See section 5.
3	Line of business has fixed choices	Fixed	15 to 135 entries, plus a free-text escape hatch.
4	GUI issue in zoning	Already fixed	Overlapping banner gone. Modal title also corrected, section 4.
5	Mayor's permit should be zoning clearance	Fixed	Business permit made implicit; Zoning Clearance added under CPDO.
6	No form presented once clicked	Already fixed	Step map shows upfront; office forms slot in on selection.

#	ITEM	STATUS	DETAIL
7	Chatbot does not work properly	Fixed	Answers the permit you named. Five distinct defects, section 3.
8	Sorting/filter in payment history	Already fixed	Verified the list reorders.
9	Birthday needs validation	Already fixed	Client and API. Kept under regression test.
10	Verify the attached image belongs on the user side	Fixed	Same finding as item 23.
11	Application dates should be auto-filled	Fixed	Derived from submission, shown read-only.
12	Other fields optional, should be required	Fixed	Registration number, TIN, registration type. Trade name left optional on purpose.
13	Allow multiple responses	Fixed	Replies become a thread. Existing replies migrated.
14	Drafts are not saved	Already fixed	Saved, reopened, every field restored.
15	No validation for TIN	Fixed	9 digits plus optional branch code; separators normalised.
16	Date issued should come from the admin side	Fixed	Issuance dates moved to the officer review screen.
17	Next button gating	Already fixed	Disabled, and names what is missing.
18	Revenue code sections must not be mentioned	Fixed	Also stripped from the API payload, not just hidden.
19	Other Requirements page locked	Already fixed	Page loads for BPLO staff.
20	Form did not reflect on the admin side	Already fixed	Office forms and fee declaration render for the officer.
21	Session security	Fixed	Portal separation plus response hardening. Section 7.
22	Notification description on admin side	Fixed	Applicant copy was being shown to officers.
23	Form asks renewal vs new	Fixed	Derived server-side; a contradicting client value is not trusted.
24	Admin login should be separate	Fixed	Two doors, enforced by the API.
25	Download Receipt does not work	Already fixed	Downloads a valid PDF with the correct reference and amount.

Items 10 and 23 turned out to be the same finding reported twice: the FSIC sheet asking which certificate is being applied for.

3. Bugs found that were not on the checklist

These surfaced from walking the flows rather than from the list. Each is the kind that passes a type-check and fails in front of a user, which is exactly the class the old verification method could not catch.

Blocking an applicant who has done nothing wrong

1. **A new application demanded a "Previous Mayor's Permit."** The requirement is correctly marked renewal-only in the database and the API sends that context, but the wizard ignored it. A brand-new business was blocked by a document that cannot exist. Fixed and verified in both directions: new applications no longer list it, renewals still do.
2. **Every file attachment from the Respond form was rejected** with a 422. The form has no document-type picker so it never sent one, while the API required it alongside the file. Silently, from the applicant's side.
3. **Renewal prefill blanked a free-text line of business**, so renewing a business whose trade is "Other" lost the text and blocked a step the applicant had completed the previous year.
4. **Office forms rejected an empty payload.** With the FSIC sheet now fully derived, the wizard would have failed on Next with nothing left to send.

Blocking an officer

5. **Recording issuance dates was unreachable.** The route required a permission officers do not hold. The feature existed and could not be used.

Telling someone something untrue

6. **The chatbot quoted fees from a superseded column.** It answered "P700 for FSIC" from the legacy flat schedule the revenue-code engine replaced. Under the ordinance, FSIC is 10% of the mayor's permit plus regulatory fees. That is a number the system will not charge, and the kind of wrong an applicant repeats back to a cashier. It no longer quotes any peso figure it cannot source from an active fee rule.
7. **The zoning modal announced "CONGRATULATIONS!"** in green for what is only a recorded map pin. The system does not evaluate zoning; the City Planning Office rules on conformance during processing. Announcing a result nobody has determined is how an applicant comes to believe their zoning passed. Retitled "Location recorded".

Infrastructure

8. **Unauthenticated API requests returned a 500, not a 401.** A tester pasting a URL into the address bar got a server error page.
9. **The chat button covered the chat panel's own send button.**

4. Where the work deliberately did not follow the instruction

Three places where doing exactly what was asked would have made the system worse or less true. Each is flagged so you can overrule it.

Trade name stays optional (item 12)

The item says other fields should be required. Registration number, TIN and registration type now are. Trade name is not, because many sole proprietors genuinely have none, and requiring it would force applicants to type something false.

Occupancy "Application Type" stays a question (items 10, 23)

The rule was: stop asking what the system already knows. On the Occupancy sheet, "Application Type" means Full versus Partial occupancy. That is occupancy scope, a real applicant decision, not the new-versus-renewal the system knows. Removing it would have been over-applying the rule.

No semi-annual payment option (item 2)

The generic national unified form offers annual, semi-annual and quarterly. Malabon's own ordinance (Sec. 2N) grants exactly two: in full within the first 20 days of January, or four instalments within the first 20 days of January, April, July and October. Offering a third would be the system inventing a payment schedule the ordinance does not allow. It is absent, and validation rejects it.

5. Item 2: what is still missing

This is the one item that cannot be closed from here. Four fields from the standard unified form were added: lessor details (only required when the premises are rented), an emergency contact, mode of payment, and how many employees live in Malabon.

What would close it: a scan or clear photo of the application form the Malabon BPLO actually hands out. Everything above is inferred from the national unified form plus the ordinance. That is a reasonable basis, but it is not the same as matching the document in use at the counter.

6. The verification gap that made this necessary

Worth stating plainly, because it explains how an earlier round of fixes was reported as done while several were broken.

The frontend type-check everyone was running is `npx tsc --noEmit`. The repository's root TypeScript config is a solution-style file containing `"files": []`, which means it compiles nothing and exits successfully regardless of what is in the source.

It has been the green light on every frontend change in this repository while checking none of them.

Two real errors reached me through that gap in this session alone, both caught only by the production build. The fix adds `npm run typecheck`, which runs the same project references the build uses. It was proved by introducing a deliberate type error: the old command exited 0 in silence, the new one failed on it.

Recommendation: point any continuous-integration job or pre-commit hook at `npm run typecheck`, not the old command. Otherwise this recurs.

7. Session security (item 21)

The tester marked this optional. What is now in place:

- **Separate sign-in doors, enforced on the server.** A staff credential is refused at the citizen sign-in and a citizen credential at the staff sign-in. The refusal happens only *after* the password has already verified, so it cannot be used to discover which email addresses are staff accounts.
- **Response hardening** on every reply: frame denial, content-type sniffing off, no referrer leakage, a restrictive content security policy, and strict transport security over TLS. The bearer token lives in browser storage, which makes cross-site scripting and clickjacking the realistic hijacking routes rather than network interception; these close both. They are set in the application rather than the proxy so they hold in development and behind a tunnel.
- **Officer and admin routes are permission-guarded in the router**, not only by the API. Defence in depth, not the defence.
- Carried over: 12-hour token expiry, revocation on password change, account lockout after five failed attempts, and a login rate limit.

Deliberately not done: binding a session to an IP address or user agent. It would break testers moving between mobile networks for very little gain against the actual threat.

8. Fee engine coverage

The permit fees are computed from 423 active rules transcribed from the New Revenue Code of Malabon (City Ordinance A10-2016), replacing a flat per-permit schedule under which a sari-sari store and a large manufacturer paid the same amount.

RULE GROUP	ACTIVE RULES	NOTES
Regulatory (sanitary, garbage, environmental, other permits)	193	Highest-rate and capping logic per the ordinance
Mayor's permit	165	Includes the catch-all, held back unless nothing specific matched
Business tax	36	Per line of business; franchise and printing route in lieu
Administrative fees	10	
On-demand services	7	
Zoning / locational clearance	4	Added this round; see the open question below
Exemption claims	3	BMBE, cooperative, petroleum
Community tax, city charges, fire code, penalties	5	Fire fee computed last, over the permit and regulatory subtotal

Where the ordinance leaves an amount to official discretion, the system emits an officer-assessed line rather than inventing a number. That covers presumptive income level assessments, market stall rentals, and the

BMBE and cooperative exemption claims.

9. Open questions for the LGU

Four things need a decision from someone with authority over the ordinance. The first is the only one that touches money directly.

Q1. Zoning clearance, Sec. 3.D.01(d): needs confirmation before real billing

The subsection prints four lines, all at ₱345.00: (1) land use verification fee, (2) processing fee, (3) sari-sari stores, (4) all other businesses. The ordinance does not state whether (3) and (4) are *additional* charges or simply restate that the same rate applies to every business category.

The system currently bills (1) and (2), which mirrors the verification-plus-processing structure the same article uses for buildings, giving **₱690.00** plus the ₱45.00 filing fee. The alternative readings give **₱345.00** or **₱1,380.00**.

This is recorded as a defect against the rule so it stays visible. Please confirm the intended reading.

Q2. Should applicants see ordinance citations at all?

Item 18 says section numbers must not be mentioned, with a note that this would be confirmed with the LGU. They are currently hidden from applicants and kept for officers, who need them to defend an assessment. Reversing it is a one-line change either way.

Q3. The real application form

See section 5. A scan closes item 2.

Q4. Fee amounts generally

The questions raised in the earlier fee engine document (Q1 to Q10 there) still stand, including the certified copy of the manufacturer bracket pages and the separate Market Code ordinance.

10. Pull requests

One pull request per problem, each with its own reasoning and evidence. They are stacked: each is based on the one before it, so a given request's diff shows only its own change. Merge in order.

PR	ITEMS	TITLE	LINK
#1	foundation	Revenue-code fee engine (Ordinance A10-2016)	github.com/antuK3n/biztrack/pull/1
#2	1	Working profile and password endpoints	.../pull/2
#3	foundation	Chatbot foundation	.../pull/3
#4	6, 8, 14, 19, 20	Apply-flow rework and admin visibility	.../pull/4
#5	21, 24	Separate staff sign-in and response hardening	.../pull/5
#6	18, 22	Hide ordinance citations; fix officer notification copy	.../pull/6
#7	13	Allow multiple responses to one requirement request	.../pull/7
#8	10, 11, 16, 23	Office forms stop asking what the system knows	.../pull/8
#9	3, 5, 12, 15	Zoning clearance, open line of business, required registration	.../pull/9
#10	7	Chatbot answers the question that was asked	.../pull/10
#11	fee data	Zoning clearance fees from Ch. III Art. D	.../pull/11
#12	2	Unified application form fields	.../pull/12
#13	tooling	Add a typecheck script that actually checks types	.../pull/13
#14	documentation	System testing checklist status	.../pull/14

11. Summary

Of the 25 checklist items, 15 were fixed this round, 8 were already fixed and re-verified, and 2 remain partly open pending the LGU: item 2 needs the paper form, item 18 needs a confirmation that is already implemented in the direction the tester asked for.

Nine additional bugs were found and fixed that were not on the checklist. Four of them would have blocked an applicant who had done nothing wrong, and two would have told someone something untrue about money or about a decision the city had not made.

The most consequential finding is not on the checklist at all: the frontend type-check that has been signing off on changes checks nothing. That is fixed, and it is worth wiring into whatever runs before a merge.

Prepared from the repository at commit stack `pr/01` through `pr/14` . Figures in this report were read from the running system, not from notes: 423 active fee rules, 135 PSIC entries, 121 passing tests, 86 changed files.